

// Technical Proposal + Pricing

5-Channel Social Media Automation

PHASE 1A Orchestration + Publishing Layer

PREPARED FOR	DATE	DOCUMENT TYPE	STATUS
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What This Proposal Covers

Right now you have two workflows that each do their job in isolation — HeyGen generates your avatar videos and Blotato handles the posting. The problem is they share no information. There's no central brain between them, so someone still has to manually wire the output of one into the input of the other every single time. That's not automation. That's task delegation with extra steps. Phase 1A solves exactly this. It builds a Google Sheets CMS that sits between both workflows — a single source of truth that everything reads from and writes back to. Once that layer exists, the machine runs itself.

— 02 · THE PIPELINE

How the System Works — End to End

01 CONTENT INPUT

A topic row is written into the Google Sheets Content Queue — either manually (Phase 1A) or generated automatically from a News API trigger (Phase 1B). Each row contains the topic, target platform, content type, and scheduled time. Nothing fires until a row exists.

Google Sheets / Content Queue

02 AVATAR VIDEO GENERATION

The HeyGen workflow reads the next ready reel row from Sheets, fires the avatar video generation with your Avatar ID and Voice ID pre-loaded, waits for completion, saves the output to Google Drive, and writes the Drive file ID back to the row. Status updates to ready_to_post.

HeyGen API Google Drive Google Sheets / Status Update

03 IMAGE GENERATION

For photo posts, the image generation workflow reads the row and routes to the correct model — Flux Pro via Replicate for LinkedIn photorealism, DALL-E 3 for Facebook and Instagram graphics. Output saves to Drive. Row status updates.

Flux Pro / Replicate API DALL-E 3 / OpenAI Google Drive

04 QUEUE LOCK — COLLISION PREVENTION

Before any publisher touches a row, the workflow sets its status to locked. This is the first action, before any API call. Even if two n8n instances wake simultaneously, the second sees locked and skips. No duplicate posts. No race conditions.

Google Sheets / Row Lock Pattern

05 PUBLISHING — ALL 5 CHANNELS

The publisher reads the locked row, confirms the Drive file is accessible, and posts to the target platform via its API. Each channel runs its own publisher node. Rate limit counters check daily caps before dispatching. On success, status updates to done with the platform's post ID logged.

TikTok Content API LinkedIn Graph API Meta Graph API YouTube Data API v3

06 ERROR HANDLING + LOGGING

On any failure, the row's attempt_count increments and status sets to retry_scheduled. Retry fires after 2 minutes, then 8, then 30. After 3 failures, status becomes failed_alert and a Telegram notification fires with the row reference. Every run appends to the Logs tab — platform, status, error message, timestamp.

n8n Wait Node Telegram Bot API Google Sheets / Logs Tab

Why This Won't Break at Volume

The fragility concern is valid. Most n8n automations break because workflows share state they shouldn't, or because failures go undetected. Here's exactly how this system prevents that.

- ✓ **Row locking prevents collisions.** The status column is the single arbiter of what's being processed. Locked rows are invisible to every other workflow instance.
- **Per-platform rate counters live in the Settings tab.** The queue checks counts before dispatching — TikTok daily limits, carousel counters per channel. You never exceed platform quotas.
- ⚠ **Failures are explicit, not silent.** Every failed row has a visible status and a logged error message. Nothing disappears into a void. You know exactly what failed and why.
- **The Logs tab gives you a full audit trail.** Every run writes a row — timestamp, platform, post ID, status, error. You have a complete history without opening n8n.

— 03 · SCOPE

What I'll Build

GOOGLE SHEETS CMS — THE ORCHESTRATION LAYER

Sheets becomes the single source of truth. Three tabs handle everything: Content Queue (one row per post), Settings (rate counters, API config, approval toggles), and Logs (full execution history). Workflows read from it and write back to it. Nothing posts unless a row says it's ready.

COLUMN	PURPOSE
topic	What the post covers
content_type	reel photo carousel
platform	TikTok LinkedIn Facebook Instagram YouTube
status	ready → locked → posting → done / failed / retry_scheduled
scheduled_time	When to post
drive_file_id	Generated media file location in Drive
post_id	Platform's confirmation ID after a successful post
attempt_count	Retry counter — max 3 before failed_alert
error_log	Last error message if the row fails

WORKFLOW ADAPTATIONS

The HeyGen workflow gets its input source swapped from hardcoded to Sheets-driven. The Blotato posting nodes are kept — only the input changes. Both workflows are adapted, not rebuilt. Estimated adaptation time: 6-10 hours depending on how they're currently structured.

CHANNELS IN SCOPE

CHANNEL	CONTENT TYPE	STATUS
TikTok	Avatar reels	Included
LinkedIn	Photos + carousels	Included
Facebook	Reels + photos + carousels	Included
Instagram	Reels + photos + carousels	Included
YouTube	Shorts	Phase 1B

INTENTIONALLY DEFERRED

FEATURE	REASON
Content generator (News API + GPT-4 + image gen)	Phase 1A uses manual Sheets input
Exponential backoff retry logic	Basic retry only — manual fix on persistent failures
Telegram daily summary	Phase 1B — error alerts only in Phase 1A
YouTube Shorts	Same architecture, added when Phase 1A is stable
Carousel counter automation	Carousels added manually to queue for v1
Approval / review mode toggle	Phase 2 feature

Without the content generator, you write topics into Sheets manually for Phase 1A. That means the 5-minutes-per-day target is not fully achieved yet — closer to 15-20 minutes daily. Phase 1B eliminates that. The trade-off is intentional: prove the publishing layer works first, then automate the content input.

— 04 · TIMELINE

Milestone Breakdown

WEEK 1

Sheets Schema + HeyGen

- Google Sheets CMS built out — Content Queue, Settings, and Logs tabs
- HeyGen workflow updated to read topic and avatar ID from Sheets
- Generated videos save automatically to Google Drive
- Row status updates correctly after each generation

Checkpoint — HeyGen generates an avatar video from a Sheets row with no manual input

WEEK 2

All Publishers Live

- Blotato channel nodes wired to Sheets queue as new input source
- Row lock pattern active — no duplicate posts possible
- Daily rate limit counters running per platform
- All 4 channels posting successfully on a manual trigger

Checkpoint — One full cycle completes — row enters queue, video generates, posts to all 4 channels, row updates to done

WEEK 3

Stability + Handoff

- End-to-end automated run — no manual trigger required
- Failed rows clearly flagged in Sheets with error details
- Cross-platform testing across all content types (reel, photo, carousel)
- Handoff documentation: Sheets schema, workflow logic, adding new content rows

Checkpoint — System completes a full automated cycle with no manual input, all 4 channels post correctly

— 05 · ONGOING COSTS

Ongoing Platform Costs

Start on n8n cloud (\$20/month). It handles Docker, SSL, and backups — no DevOps overhead while the workflows are being proven. Once Phase 1A is stable, migration to a self-hosted VPS (Hetzner CX21, €5/month) takes 30 minutes: export workflows as JSON, import on the new instance, done.

SERVICE	MONTHLY ESTIMATE
HeyGen (60 avatar videos/month)	\$80 – \$120
Flux Pro via Replicate (LinkedIn photos, 90/month)	~\$5
DALL-E 3 (Facebook + Instagram photos)	~\$7
n8n cloud	\$20
Total operational cost	~\$112 – \$152 / month

— 06 · INVESTMENT

Investment

Recommended starting point	\$1,200 – \$1,500	Phase 1A — 4 channels, 3 weeks
Add-on after Phase 1A stability	\$800 – \$1,000	Phase 1B — content gen, retry, Telegram, YouTube
Full system	\$2,000 – \$2,500	Phase 1A + 1B combined

The Phase 1A range depends on the condition of your existing workflows. Once I see them — JSON export from n8n or a 2-minute Loom of the canvas — I'll give you a single fixed number.

PAYMENT TERMS

50% upfront to begin · 50% on handoff once the automated cycle runs clean across all 4 channels.

— 07 · NEXT STEP

What I Need From You

- Export your existing workflows from n8n as JSON — or record a 2-minute Loom of the canvas
- Share your HeyGen Avatar ID and Voice ID
- Confirm API access: Google Sheets, Google Drive, TikTok, LinkedIn, Facebook, Instagram
- Confirm n8n cloud account or VPS preference